

Tristan Lazard

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*Machine learning researcher with expertise in **self-supervised learning** and **representation learning** from complex, high-dimensional data. My work focuses on building models that learn meaningful structure from **raw, unlabelled inputs**, from gigapixel microscopy images to multi-omics, and on developing **interpretability** methods to uncover the world knowledge these representations encode.*

PROFESSIONAL EXPERIENCE

Microsoft Research

Researcher

Cambridge, UK

June 2025 – Present

Research at the intersection of biology and AI within the Health Futures team, spanning two complementary directions:

- **Interpretability for discovery:** developing methods and agentic workflows that expose the knowledge encoded by large models and turn model predictions into testable biological hypotheses, spanning sparse concept attribution, automatic interpretation of visual concepts, and automated histopathology analysis ([SCOPE](#), [visual concepts](#), [NOVA](#)).
- **Multimodal training:** investigating how to mix heterogeneous data in large autoregressive models, measure synergy between modalities, and promote compositionality.

SynagenAI

Software Developer

Dresden, Germany

Oct 2024 – April 2025

Early-stage startup building agentic AI products for healthcare. Helped shape the company's first prototypes, notably a tumor-board decision-support system and a research tool for clinical trial matching.

CBIO – Mines-Paristech

PhD in BioInformatics 📖 📄

Paris, France

Dec 2019 – Dec 2023

- Developed interpretability methods for Multiple Instance Learning (MIL) architectures to uncover morphological patterns of Homologous Recombination Deficiency (HRD) in breast cancers [\[paper\]](#).
- Created **GigaSSL**, the first self-supervised learning algorithm operating at the whole-slide scale (one slide → one representation), matching or outperforming specialised supervised methods across a wide range of downstream tasks ([paper 1](#), [paper 2](#)).
- Validated GigaSSL in international data challenges: DrivenData-VisioMel (1st/542 public – 3rd/542 private, \$5,000 cash prize), Kaggle-UBC-Ocean (2nd/1326 public – 17th/1326 private). A simple linear model on GigaSSL representations competed with complex MIL and ensembling pipelines.

Achievements:

- **5 publications** (3 first-author) in biological journals and machine learning venues. 1 patent.
- **Prize-winning solution (\$5,000)** at the [VisioMel challenge](#) [\[code\]](#).

CMM – Mines-Paristech

Engineering Internship

Fontainebleau, France

Jan – July 2019

Automatic counting of melanocytes: In partnership with L'Oréal, developed a deep-learning algorithm for automatically counting melanocytes in fluorescence microscopy images. The solution was integrated into the company's processing pipeline and led to a peer-reviewed publication.

IBPS – Sorbonne University

Research Internship

Paris, France

Jan – Aug 2018

Large-scale phylogenetic tree inference: Developed an algorithm for reconstructing phylogenetic trees from large-scale genomic sequence data.

PUBLICATIONS

First-author publications

Lazard, T. et al. (2026). Sparse concept attribution for histomorphological hypothesis generation from whole-slide classifiers. [arXiv preprint](#).

Lazard, T.*, Beaufrière, A.* et al. (2025). Self-supervised learning to predict intrahepatic cholangiocarcinoma transcriptomic classes on routine histology. **JHEP Reports**.

Lazard, T. et al. (2024). Democratizing computational pathology: optimized Whole Slide Image representations for The Cancer Genome Atlas. **Preprint**.

Lazard, T. et al. (2023). Cancer du sein – Utilisation de l’intelligence artificielle pour prédire le statut tumoral relatif à la recombinaison homologue. **Médecine et Sciences**.

Lazard, T., Lerousseau, M., Decencière, E., Walter, T. (2023). Giga-SSL: Self-Supervised Learning for Gigapixel Images. **CVPR Workshops**.

Lazard, T. et al. (2022). Deep Learning identifies new morphological patterns of Homologous Recombination Deficiency in luminal breast cancers from whole slide images. **Cell Reports Medicine**.

Lazard, T. et al. (2022). Applying Deep Learning to Melanocyte Counting on Fluorescent TRP1 Labelled Images of In Vitro Skin Model. **Image Analysis & Stereology**.

Co-authored publications

Viart, N. M., Thibault, L., **Lazard, T.** et al. (2026). Deep learning analysis of breast cancer histology predicts ATM pathogenic variant carrier status. **npj Digital Medicine**.

Meissen, F., Bouzid, K., **Lazard, T.** et al. (2026). Automatic Interpretation of Visual Concepts. **ICLR Workshop on Unifying Concept Representation Learning**.

Vaidya, A. J., Meissen, F., Castro, D. C., Bannur, S., **Lazard, T.** et al. (2025). NOVA: An Agentic Framework for Automated Histopathology Analysis and Discovery. **arXiv preprint**.

Lubrano, M., **Lazard, T.** et al. (2022). Automatic Grading of Cervical Biopsies by Combining Full and Self-supervision. **ECCV Workshops**.

Naylor, P., **Lazard, T.** et al. (2022). Prediction of Treatment Response in Triple Negative Breast Cancer From Whole Slide Images. **Frontiers in Signal Processing**.

** Equal contribution.*

EDUCATION

PSL University <i>ITI – MS Innovation, Entrepreneurship and Innovation</i>	Paris 2018 – 2019
Sorbonne University <i>Master 2: Applied Mathematics</i>	Paris 2016 – 2017
Thesis: “ <i>A model of random mutualistic network.</i> ”	
École Normale Supérieure <i>Master 1: Evolutionary Biology and Ecology / Bachelor in Life Sciences</i>	rue d’Ulm, Paris 2013 – 2016
Lycée Saint-Louis <i>Classe préparatoire BCPST</i>	Paris 2011 – 2013

SKILLS

Languages	French (native), English (fluent), Spanish (beginner)
Programming	Python, Bash, L ^A T _E X, Nextflow
Tools & Infra	Docker, Slurm, cloud computing, GPU computing, PyTorch, OpenCV, scikit-learn

INTERESTS

Climbing, harmonica, poetry and literature.